

Etsy AI-Generated Image Detection: Leveraging EVA-02 Vision Transformers and Focal Loss for Robust Deepfake Classification

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Abstract—Image detection of AI-generated images is an increasingly significant issue as generated models create synthetic images of increased photorealism. In this paper, we have a binary image classification system that separates the images created by AI and real-world images by a three-part upgrade on top of a ConvNeXt-Small baseline. To reduce the resolution to 448x448 first, the backbone is substituted by EVA-02 (eva02_base_patch14_448), a Vision Transformer, which adds images at 448x448 resolution and applies self-attention on the entire image to identify inconsistencies in the structure. Second, we change Focal Loss (loss = 2.0, and 0.25) instead of Binary Cross-Entropy loss to bias easy examples and focus learning on hard and photorealistic cases. Third, JPEG compression simulation and Gaussian noise injection are added to the augmentation pipeline so that low-level pixel artefacts are discouraged, structural reasoning encouraged. Training AdamW with differential learning rates and CosineAnnealingLR schedule, ten epochs, and a one-epoch warm-up phase to preserve the pre-trained backbone weights. The system has an F1 score of 0.9144 on a held-out validation set of 960 images with balanced precision and recall of 0.92 on each of the two classes.

Index Terms—AI-generated image detection, Vision Transformer, Focal Loss, EVA-02, synthetic imagery.

I. INTRODUCTION

A. Problem Definition

Generative neural networks such as Generative Adversarial Networks (GANs), diffusion models, and large multimodal systems have become a rapidly expanding category of generative artificial intelligence systems. They are now capable of creating images that are more real and than before. E-commerce platforms such as Etsy need merchants to use actual photographs of their good, and so, fake images created by AI are a major issue regarding authenticity and trust. So, automated detection systems need to be able to tell on a large scale whether a picture of a product was taken with a camera or made by a generative model.

This dichotomous grouping endeavor is not simple. Earlier generative algorithms produced artifacts that would be discovered via basic texture or frequency-domain detection. Diffusion-based generators are currently capable of producing images that have the right global lighting, lifelike material

properties and consistent geometry on a scene. The disjunctive cue between the two groups becomes increasingly identified on minor structural variations, such as anatomically implausible hand poses, unnatural symmetry in background features, or overly parallel textile surface. In this way, a detection model should consider the entire image structure rather than merely considering noise signatures on which it has been trained.

B. Baseline Limitations

The previous baseline used a ConvNeXt-Small, architecture as implemented in the shape of 224 at 224 pixels and with a Binary Cross-Entropy (BCE) loss. Although ConvNeXt-Small is an effective convolutional network to solve general image classification tasks, it has two limitations here.

Receptive Field Constraints: Convolutional networks are a sequence of local operations on images. Although the large effective receptive fields are found in the deeper layers, global structural relationships such as the relative position of the anatomical landmarks in a complete image are deduced indirectly. Vision Transformers self-attention models realize a pairwise association between each image patch directly in a global context, and thus they provide access to such information in the very first layers.

Loss Function Imbalance: Standard BCE loss is the one that assigns the same weight to each training sample. On a set of cartoons that contain clearly artificial cartoons, as well as photorealistic deepfakes, the model gradient is crowded with a significant number of easy examples. The informative, hard cases that actually represent the complexity of the challenge, manipulated photographs under the radar, give relatively little contribution to parameter updates to restrict how the model may fine-tune on the most significant examples.

C. Proposed Strategy

The present work deals with the limitations identified in terms of the three-component strategy of upgrading. The architecture is substituted with EVA-02 [fang2023] which is a high-resolution Vision Transformer. Focal Loss [lin2017]

substitutes the loss function, to focus learning on hard examples. The augmentation pipeline is augmented to disinterspers the use of artifactual noise signatures. The combination of these changes creates a system that rationates concerning the structural and not the low-level pixel distributions of images.

II. DATASET AND DATA PIPELINE

A. Dataset Composition

The dataset is a total of 6,858 images (extracted) of a challenge archive (genai_image_challenge.zip). Among them, 4,800 images are provided with ground-truth to train on them and 2,058 images are an unlabeled test set to final inference and submission. The distribution of labels in the training set is almost equal: 2,485 images are labelled as authentic (class 0) and 2,315 images are labelled as AI-generated (class 1) with a class ratio of about 52:48. This near-parity ensures minimal occurrence of extreme class imbalance, the Focal Loss formulation endures residual imbalance using its parameter, the weighting parameter, termed alpha.

A preprocessing step was used to eliminate missing files, which protects against runtime errors that otherwise would stop training. Images not found in the filesystem were determined and excluded out of the dataframe before the instantiation of the dataset. This safety step will prevent the DataLoader from trying to read a file that is not there and will never add tensors of zero fillings to the training gradient.

B. Train/Validation Split

The 4,800 images with labels were divided into a training set of 3,840 images (80 percent) and a validation set of 960 images (20 percent) with scikit-learns train test split. Among them, 4,800 images have ground-truth labels to use in supervised training, which is to be followed by 2,058 images constituting an unlabelled test set to perform final inference and submit it. The distribution of labels in the training set is nearly balanced: 2,485 pictures are real (class 0) and 2,315 pictures are AI-generated (class 1), which creates a class ratio of approximately 52:48. This close to equal distribution lessens worries about a big class imbalance. However, the Focal Loss formulation’s weighting parameter takes care of any remaining imbalance..

C. Augmentation Strategy

All the changes were made using the Albumentations library as it has NumPy-based transformations that are efficient on a GPU. Table I is used sequentially to add the augmentations to the training pipeline

1) *Rationale for ImageCompression and GaussNoise:* ImageCompression and GaussNoise should be given special attention since these augmentations are driven by a particular hypothesis suggesting the failure mode of the baseline model. Generative models, especially GAN-based and diffusion-based models, introduce characteristic high-frequency artefacts to their outputs: periodic structure in frequency space (GAN spectral peaks), particular dithering structure (DALL-E), or particular grain structure (Stable Diffusion).

TABLE I
AUGMENTATION PARAMETERS AND RATIONALE

Augmentation	Parameters	Rationale
Resize	448 × 448 px	Matches EVA-02 native input resolution.
HorizontalFlip	$p = 0.5$	Standard geometric invariance; AI images are horizontally symmetric.
VerticalFlip	$p = 0.2$	Low-probability vertical inversion for additional geometric diversity.
RandomBright.	$\pm 20\%$, $p = 0.4$	Simulates variation in lighting and camera sensor response.
Affine	Scale 0.9–1.1, rot $\pm 15^\circ$	Simulates perspective variation and prevents positional overfitting.
ImageCompress.	Qual. 60–99, $p = 0.5$	Simulates JPEG compression artefacts. Prevents memorising high-frequency noise.
GaussNoise	Var 10–50, $p = 0.4$	Injects random Gaussian pixel noise. Forces model to ignore low-level statistics.
Normalize	ImageNet μ/σ	Zero-centres pixel values to the pre-trained distribution.

A neural network that has not been trained to induce noise can be trained to learn to identify these particular artefacts over structural inconsistencies that characterize AI-generated imagery in a domain-agnostic way. The augmentation pipeline is trained to corrupt exactly the signal that a naive model could over-fit. The augmentation pipeline introduces JPEG compression (which kills high-frequency information) and Gaussian noise (which conceals fine pixel-level statistics) in random locations to corrupt the signal that the naive model may be over-fitting.

The classifier is thus forced to base its predictions on features that resists such degradations i.e., mid-level and high-level structural features: implausible anatomical structures, unnaturally perfect textures, inconsistent depth of field, and aberrant shadow geometry. These are the very qualities that demonstrate the underlying drawbacks of existing generative models. Validation transform does not use any augmentations, since augmentations during evaluation would add randomness to the evaluation metric, and the evaluation metric cannot be compared across epochs.

III. METHODOLOGY AND ARCHITECTURE

A. Vision Transformer: EVA-02

1) *Background and Selection:* The architecture chosen to use in this system is EVA-02 in its basic configuration: `eva02_base_patch14_448.mim_in22k_ft_in22k_in1k`. EVA-02 is a Vision Transformer (ViT) architecture that is pre-trained using Masked Image Modelling (MIM) on ImageNet-22k (about 14 million images in 21,841 categories),

and then fine-tuned on ImageNet-22k and ImageNet-1k. The model is available as the library named `timm` and has around 86.3 million parameters.

The core architectural difference of the ConvNeXt-Small baseline is the self-attention mechanism. In a Vision Transformer, the input image is divided into non-overlapping patches of size 14×14 pixels. The patches are progressively coded into a token embedding. Multi-head self-attention is then performed on all tokens with the aid of a stack of Transformer encoder layers. The query, key and value attention operation is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where d_k is the major dimension to scale with. This formulation calculates a weighted average of all values vectors, where the weighting is based upon the similarity of each query and all keys. Importantly, the attention weights are calculated on a global basis among all patch tokens enabling the model to directly correlate any two patches irrespective of their spatial distance. To obtain a similar long-range relationship, a CNN needs many convolutional layers to spread information, where the information becomes weaker at each step.

2) *High-Resolution Processing at 448×448* : The choice of a 448×448 pixel input resolution, double the 224×224 standard used by the baseline, provides four times as many pixels of information to the network[cite: 78]. With a patch size of 14×14 , the image is tokenised into $(448/14)^2 = 1,024$ patch tokens[cite: 79]. This high token count captures fine-grained local details that are invisible at lower resolutions: subtle blurring at object boundaries, characteristic pixel-value gradients at AI-generated image edges, micro-texture uniformity in hair or fabric, and the imperceptible doubling or misalignment of fine structures—such as individual teeth or eyelashes—that current generative models frequently introduce.

Computational cost is the primary trade-off for high-resolution processing. At 448×448 , each forward pass through the self-attention layers operates on 1,024 tokens, and the memory complexity of standard self-attention scales quadratically with the number of tokens, expressed as $\mathcal{O}(n^2)$. This dictates the selection of a batch size of 16, which was determined to fit within the memory constraints of a T4 or A100 GPU while maximising throughput.

3) *Global Self-Attention for Structural Anomaly Detection*: The main benefit of global self-attention when applied to AI image detection is that it has direct access to structural relationships. Take the case of searching an AI-generated six-finger hand. Before a representation of the entire hand structure occurs, a CNN would have to combine local patch information across many layers in a hierarchy. Vision Transformer, in turn, can build up an attention pattern that directly connects the patches on the fingers-tips to one another and to the wrist, and calculates structural consistency in a single attention layer. In a similar manner, an attention head can learn to attend to both eyes simultaneously in a face detection

context to confirm bilateral symmetry, which also requires explicit long-range reasoning.

This architectural benefit is backed by empirical evidence in the training logs. Validation F1 started at 0.6820 at the beginning of the warm-up epoch and grew slowly but surely to 0.9144 at epoch 10, and steadily improved with further epochs, indicating that the model was gradually learning to be more discriminative in its structural representations instead of overfitting to low-level cues.

B. Focal Loss

1) *Limitations of BCE Loss*: Standard Binary Cross-Entropy (BCE) loss for a single example with true label $y \in \{0, 1\}$ and predicted probability p is:

$$\text{BCE}(y, p) = -[y \cdot \log(p) + (1 - y) \cdot \log(1 - p)] \quad (2)$$

Standard Binary Cross-Entropy (BCE) loss assigns equal weight to each sample in the loss computation. In a dataset containing obvious synthetic cartoons (easy negatives, high confidence correct predictions) alongside subtle photorealistic deepfakes (hard positives, low confidence or incorrect predictions), the easy examples dominate the gradient during mini-batch updates. The model may achieve a globally low BCE loss while remaining systematically wrong on the hardest and most informative cases.

C. Focal Loss Formulation

To address this issue of sample imbalance, Lin *et al.* proposed Focal Loss in the context of dense object detection. It addresses this by introducing a modulating factor that down-weights easy examples:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (3)$$

where p_t is the model's estimated probability for the ground-truth class, γ is the focusing parameter, and α_t is a class-balancing weight. The key insight is the term $(1 - p_t)^\gamma$. When an example is confidently classified correctly, $p_t \approx 1$ and $(1 - p_t)^\gamma \approx 0$ effectively zero out its contribution to the loss. When an example is misclassified ($p_t \approx 0$), the modulating factor approaches 1, and the loss is equivalent to standard BCE. This asymmetry means the total gradient is dominated by misclassified hard examples.

D. Parameter Selection

The parameters were selected following the recommendations of Lin *et al.*:

- $\gamma = 2.0$: This is the standard value used in the RetinaNet detection framework. At $\gamma = 2.0$, an example classified with $p_t = 0.9$ (easy) receives a loss weight of $(1 - 0.9)^2 = 0.01$, reducing its contribution by a factor of 100 relative to a hard example with $p_t = 0.5$, which receives $(1 - 0.5)^2 = 0.25$. This effectively restricts the model's attention to examples where it is uncertain or incorrect.

- $\alpha = 0.25$: The alpha parameter provides a mild down-weighting of the majority class. Since the dataset is near-balanced (52:48), this primarily serves as an additional safety mechanism to prevent the model from slightly favouring the authentic (majority) class. In highly imbalanced datasets, α would be set closer to the minority class frequency.

IV. EXPERIMENTAL SETUP AND TRAINING STRATEGY

A. Hyperparameter Configuration

The core training hyperparameters are detailed in Table II.

TABLE II
HYPERPARAMETER CONFIGURATION

Hyperparameter	Value	Justification
Image Resolution	448×448 px	EVA-02 native res; 4× more detail than 224 px.
Batch Size	16	Max feasible for 448 px inputs on GPU.
Total Epochs	10	Sufficient for ViT convergence.
Head LR	1e-3	High rate for initialised classification head.
Backbone LR	2e-5	Low rate to fine-tune pre-trained weights.
Weight Decay	0.01	L2 regularisation; AdamW weight decay [4].
Warm-up Duration	1 epoch	Backbone frozen for epoch 1.
Optimiser	AdamW	Prevents interaction with scaling.
LR Scheduler	CosineAnneal.	Smooth cosine decay to eta_min = 1e-7 [5].
Loss Function	Focal Loss	$\gamma = 2.0, \alpha = 0.25$; focuses on hard examples.
Mixed Precision	AMP	bfloat16/float16 for faster throughput [8].

B. Differential Learning Rate Strategy

A critical engineering decision in this experiment is the use of two distinct learning rates applied simultaneously to different parameter groups. The final classification head (a single linear layer mapping from the EVA-02 feature dimension to a single output logit) is randomly initialised and must learn the classification function from scratch. Applying a high learning rate ($LR_{\text{HEAD}} = 1 \times 10^{-3}$) to this layer enables rapid adaptation.

The backbone parameters, however, encode rich visual representations acquired from pre-training on ImageNet-22k. Applying a high learning rate to these parameters would cause large gradient updates that overwrite the pre-trained feature hierarchy—a phenomenon known as catastrophic forgetting. By setting the backbone learning rate to 2×10^{-5} (approximately 50 times smaller), parameter updates are small enough to refine the pre-trained representations toward the specific features relevant for AI image detection without destroying the general-purpose visual knowledge.

C. Warm-Up Phase and Backbone Freezing

The warm-up strategy further protects the pre-trained representations. During the first epoch, all backbone parameters have their gradient computation disabled. Only the classification head is trained. This ensures that the head arrives at a stable initial mapping before the backbone weights are exposed to gradient updates. Backpropagating noisy gradients from a randomly initialised head through the backbone would update the pre-trained representations in a random direction, potentially damaging the feature hierarchy.

D. Learning Rate Schedule

CosineAnnealingLR decays each parameter group’s learning rate following a cosine curve from its initial value to $\eta_{\min} = 1 \times 10^{-7}$ over $T_{\max} = 10$ epochs. The cosine schedule provides a warm learning rate for the early epochs and a cold learning rate in the final epochs.

E. Mixed Precision Training

Automatic Mixed Precision (AMP) was enabled through `torch.amp.GradScaler` and `torch.amp.autocast`. During the forward pass, matrix multiplications are executed in `float16` or `bfloat16` precision, approximately halving memory consumption and doubling throughput on NVIDIA tensor cores.

TABLE III
EPOCH-BY-EPOCH TRAINING PROGRESS

Ep.	Loss	Val F1	Best?	Pred Dist (AI/Tot)
01	0.0641	0.6820	Yes ★	320 / 960 (prob [0.05, 0.88])
02	0.0563	0.7927	Yes ★	362 / 960 (prob [0.01, 0.96])
03	0.0359	0.7886	—	346 / 960 (prob [0.00, 0.98])
04	0.0256	0.8506	Yes ★	414 / 960 (prob [0.00, 0.99])
05	0.0170	0.8943	Yes ★	445 / 960 (prob [0.00, 0.99])
06	0.0101	0.8788	—	420 / 960 (prob [0.00, 0.99])
07	0.0069	0.9061	Yes ★	506 / 960 (prob [0.00, 1.00])
08	0.0051	0.9030	—	434 / 960 (prob [0.00, 1.00])
09	0.0038	0.9136	Yes ★	463 / 960 (prob [0.00, 1.00])
10	0.0030	0.9144	Yes ★	460 / 960 (prob [0.00, 1.00])

F. Epoch-by-Epoch Training Progress

The model was trained for ten epochs with best-checkpoint saving based on Validation F1 Score. Table III presents the training loss and validation F1 for each epoch.

The transition from epoch 1 (warm-up, backbone frozen, $F1 = 0.6820$) to epoch 2 (backbone unfrozen, $F1 = 0.7927$) represents the largest single-epoch improvement (+0.1107), confirming that fine-tuning the backbone parameters is critical to performance. Training loss decreases monotonically from 0.0641 to 0.0030, with no evidence of gross overfitting, as validation F1 continues to improve throughout.

G. Final Validation Metrics

The best-performing checkpoint (epoch 10) was loaded for final evaluation. The classification report computed on the full 960-image validation set is shown in Table IV.

The system achieves a Final Validation F1 Score of 0.9144, surpassing the challenge target of $F1 > 0.90$. Performance is symmetric across classes. The marginal 0.01 recall deficit for the AI-generated class is consistent with the slight majority of authentic images in the validation set and the Focal Loss $\alpha = 0.25$ parameter providing mild down-weighting of the minority class gradient.

H. Test Set Inference

Inference was performed on all 2,058 test set images using the best-checkpoint model in evaluation mode. The test set yielded 975 images classified as AI-Generated (47.4%) and 1,083 images classified as Authentic (52.6%), proportions consistent with the training set class distribution. Predictions were saved as a CSV submission file (submission_eva02.csv).

TABLE IV
FINAL VALIDATION METRICS

Class	Prec.	Recall	F1	Support
Authentic (0)	0.92	0.92	0.92	497
AI-Gen (1)	0.92	0.91	0.91	463
Macro Avg	0.92	0.92	0.92	960
Weighted Avg	0.92	0.92	0.92	960
Overall Acc.			0.92	960

V. DISCUSSION AND FUTURE WORK

A. Analysis of Results

The system gets a Final Validation F1 Score of 0.9144, which is higher than the challenge target of $F1 > 0.90$. All classes have the same performance. The small 0.01 recall deficit for the AI-generated class is in line with the fact that the validation set has a slight majority of real images and that the Focal Loss parameter $\alpha = 0.25$ slightly down-weights the gradient of the minority class.

B. Potential Failure Modes

The images produced by new models in the generative models that are not in the training distribution can have distribution shifts causing a reduction in detection reliability. Also, adversarially-generated AI images that are meant to match the real image statistics may take advantage of the same channel the strategy of augmentation was meant to bridge.

C. Future Work

1) *Test-Time Augmentation (TTA)*: This method creates many augmented images of all the test images and averages the model prediction. Since the F1 is 0.9144, and TTA might increase performance up to 0.92, SPT will likely be above this value.

2) *Cutout Regularisation*: The training process uses cutout to randomly block square patches in the input image, and therefore the model is trained to give predictions based on a partially masked input. Cutout regularisation is especially compatible with Vision Transformers, since the patch tokenisation automatically includes a boundary which is compatible with the augmentation.

3) *Architectural Scaling*: The notebook recognizes that in resource-constrained deployment situations, the alternative would be to use `eva02_small_patch14_336`. This reduced model works at 336×336 pixel sizes and has fewer parameters, decreasing training and inference latency.

4) *Extended Training*: The convergence was not evident in the validation F1 curve after epoch 10. It might be possible to further improve by extending the training to 15 epochs and keeping the same `CosineAnnealingLR` schedule. Vision Transformers have been known to have longer training in comparison with CNNs.

VI. CONCLUSION

The presented system for detecting AI-generated images (based on the EVA-02) met the main target of the DCU 2026 Machine Learning Challenge, reaching the desired performance level. Replacing a regular convolutional baseline with a high-resolution Vision Transformer (ViT), where all parts are treated as a global memory, the model was able to reason on the global structure of the image to detect fine-grained anatomical and geometric inconsistencies that constitute the current image generation of AI. Integration of Focal Loss was also helpful in honing the discriminative capacity of the model, since it helped effectively focus the model on training on more challenging photorealistic deepfakes, as opposed to the easier-to-train synthetic samples. In addition, the variants of JPEG compression and Gaussian noise in the augmentation pipeline were strategically selected to make the classifier strong to platform-specific artifacts instead of structural features or semantic features. Having a final Validation F1 Score of 0.9144 and balanced precision/recall on each of the two classes, this system offers a dependable and scalable system to ensure authenticity on e-commerce websites such as Etsy.

VII. CONTRIBUTIONS

The implementation, technical research, and reporting for this project were divided as follows:

Gaurang Kumbhar

- **Architectural Research and Selection**: Assessed the EVA-02 base patch14 architecture and the reasons to scale up the ConvNeXt-Small baseline to a Transformer-based model.
- **Engineering Loss Functions**: Expressed the Focal Loss equation and tested using the parameter values of $\gamma = 2.0$ and $\alpha = 0.25$.
- **Data Pipeline Authoring**: Wrote the logic that used missing images as a filter, and created the stratified

train/validation split to guarantee unprejudiced performance estimation.

- **Result Analysis:** Carried out the final evaluation on the validation set, which produced the classification report and the confusion matrix.

Aditya Dhayagude

- **Augmentation Strategy:** Implemented and trained the state-of-the-art augmentation pipeline with the `albumentations` library, using JPEG compression and Gaussian noise to avoid artifact memorization.
- **Training Optimization:** Added the Differential Learning Rate strategy to use different learning rates when training the backbone and the classification head, where the latter was used to avoid catastrophic forgetting with the former set to 2×10^{-5} .
- **Experiment Management:** Tuned the `CosineAnnealingLR` learning rate schedule and the single warm-up step to ensure that pre-trained weights were not harmed during early training.
- **Technical Documentation:** Wrote the Technical Report and the final Inference Pipeline to produce submission-ready test set predictions.

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